



Module 6: Signals, Emissions & Safety

Fourier Analysis · Modulation · Digital Signals · Spread Spectrum · RF Safety
2024-2028 Amateur Extra Question Pool



Module 6 Overview

- 5 exam questions — E8A-D (4 Qs) + E0A (1 Q) — only 10% of exam
- Lighter module — good recovery after Module 5
- But: E8 has real formula content
- → RMS voltage: $V_{rms} = 0.707 \times V_{peak}$
- → FM modulation index: peak deviation / modulating frequency
- → Carson's rule: $BW = 2 \times (\text{deviation} + \text{max audio frequency})$
- → Baud rate vs bit rate relationship
- E0A safety: only 1 question but a guaranteed point
- → Know: MPE limits, duty cycle, BeO hazard, controlled vs uncontrolled



Part 1: Fourier Analysis and Power

Subelement E8A — 1 exam question



Fourier Analysis and Harmonics

- Fourier's theorem: any complex periodic waveform = fundamental + harmonics
- Harmonics: integer multiples of the fundamental frequency
- → 2nd harmonic: $2\times$ fundamental; 3rd: $3\times$; etc.
- Square wave: fundamental + all ODD harmonics (3rd, 5th, 7th...)
- → This is why digital circuits radiate harmonics across the spectrum
- → And why switching amplifiers MUST have output low-pass filters

- RMS Voltage (sine wave):
- $V_{rms} = 0.707 \times V_{peak}$
- → Or: $V_{rms} = V_{peak} / \sqrt{2}$
- → Example: 200V peak → 141.4V RMS



Part 2: FM Modulation

Subelement E8B — 1 exam question



FM Modulation Index and Carson's Rule

- Modulation index: peak frequency deviation / modulating frequency
- → Example: 5 kHz deviation, 1 kHz audio → index = 5
- → Higher index = more sidebands = more bandwidth

- Deviation ratio: max frequency deviation / max modulating frequency
- → Design parameter (uses maximum values, not instantaneous)

- Carson's Rule — FM bandwidth:
- $BW = 2 \times (\text{max deviation} + \text{max modulating frequency})$
- → Example: 5 kHz deviation, 3 kHz audio → $BW = 2 \times (5 + 3) = 16 \text{ kHz}$
- → This gives the bandwidth containing ~98% of the signal power



Part 3: Digital Signal Theory

Subelement E8C — 1 exam question



Digital Modulation and Error Correction

- Shannon's theorem: maximum data rate depends on bandwidth AND signal-to-noise ratio
- Forward Error Correction (FEC):
 - → Adds redundant bits at the transmitter
 - → Receiver can DETECT and CORRECT errors without retransmission
 - → Used when retransmission is impossible: EME, broadcast, satellites
- Automatic Repeat Request (ARQ):
 - → Receiver detects errors and REQUESTS retransmission
 - → Used when two-way links exist: Winlink, PACTOR
- Baud rate vs bit rate:
 - → Baud rate = symbols per second (signal state changes per second)
 - → Bit rate = baud rate $\times$ bits per symbol



Modulation Schemes and Constellation Diagrams

- BPSK: 2 phase states (0° , 180°) — 1 bit per symbol; most robust
- QPSK: 4 phase states (0° , 90° , 180° , 270°) — 2 bits per symbol
- QAM: varies both amplitude AND phase
- → 16-QAM: 4 bits per symbol (16 unique states)
- → 64-QAM: 6 bits per symbol (64 unique states — more susceptible to noise)
- OFDM: many narrow subcarriers simultaneously
- → Each subcarrier uses QAM or QPSK
- → Used in DVB-T (amateur ATV), Wi-Fi, LTE

- Constellation diagram:
 - → Plots I (in-phase) vs Q (quadrature) — shows all possible signal states
 - → Each point = one symbol (specific amplitude + phase combination)
 - → Noise causes points to spread — if adjacent points overlap, bit errors occur



Part 4: Keying Defects and Spread Spectrum

Subelement E8D — 1 exam question



Keying Defects and Spread Spectrum

Keying Defects

- Key clicks: rise/fall time too fast → wide sidebands
- → Sounds like 'thwack' on nearby frequencies
- → Fix: slow rise/fall times with RC shaping
- Chirp: oscillator frequency changes on keying
- → Power supply voltage pulling the VFO
- → Fix: better voltage regulation
- Overmodulation (digital): excessive drive
- → Splatter on adjacent frequencies
- ASCII: 7-bit code (128 characters)
- Baudot (RTTY): 5-bit with shift characters

Spread Spectrum

- DSSS (Direct Sequence):
- → Multiply signal by pseudo-random code
- → Spreads signal across wide bandwidth
- → Looks like noise to others
- FHSS (Frequency Hopping):
- → Rapidly hop between frequencies
- → Using pseudo-random sequence
- Both: resistant to interference and jamming
- FCC: SS only above 222 MHz in Amateur Service



Part 5: RF and Electrical Safety

Subelement E0A — 1 exam question



RF Safety Fundamentals

- RF biological effect: thermal — RF energy heats body tissue like a microwave oven
- MPE (Maximum Permissible Exposure): FCC RF field strength limits
 - → Different limits for different frequency ranges
 - → Lower frequency = higher MPE (body absorbs less)
- Duty cycle reduces average exposure:
 - → 50% duty cycle = half the average exposure
 - → Pulsed modes (SSB, WSPR) have lower duty cycle than CW or FM
- Controlled environment: amateur operators who know about RF and take precautions
- Uncontrolled environment: general public, neighbors — STRICTER MPE limits
 - → Your neighbors are in an uncontrolled environment
 - → You must meet the stricter limit at their location



Hazardous Materials and Grounding

- Lead solder:
 - → Toxic if ingested — wash hands after soldering
 - → Use fume extraction or solder outdoors — don't breathe solder fumes
- Beryllium oxide (BeO):
 - → Found in some old RF power transistors and magnetrons
 - → Appears as white ceramic inside some transistors
 - → EXTREMELY toxic as dust — DO NOT sand, grind, or break
 - → If you find old transistors with white ceramic insulator — handle with extreme care
- CRT high voltage:
 - → Capacitors retain lethal charge even when unplugged — discharge before working



Key Numbers — Module 6

RMS voltage (sine wave)	$V_{rms} = 0.707 \times V_{peak}$
PEP measurement	At highest amplitude point; $(V_{rms})^2/Z$
FM modulation index	Peak deviation / modulating frequency
Deviation ratio	Max deviation / max modulating frequency
Carson's rule	$BW = 2 \times (\text{deviation} + \text{max audio freq})$
OFDM	Many narrow subcarriers; each QAM or QPSK
FEC	Error correction without retransmission
ARQ	Requests retransmission on error
Baud rate	Symbols per second
QPSK bits per symbol	2 bits per symbol (4 phase states)
QAM	Varies both amplitude and phase
Key clicks caused by	Rise/fall time too fast
Chirp caused by	Power supply pulling VFO on keying
DSSS	Pseudo-random code spreads signal
FHSS	Pseudo-random frequency hopping



Checkpoint Quiz 1 / 10

What is the RMS voltage of a 200V peak sine wave?

- A. 200 V
- B. 141.4 V
- C. 100 V
- D. 282.8 V

Answer: B — $V_{rms} = 0.707 \times 200 = 141.4V$



Checkpoint Quiz 2 / 10

What is the FM modulation index with 6 kHz deviation and 2 kHz audio?

- A. 12
- B. 3
- C. 0.33
- D. 8

Answer: B — Index = deviation/audio freq = 6000/2000 = 3



Checkpoint Quiz 3 / 10

Using Carson's rule, what is the bandwidth with 5 kHz deviation and 3 kHz audio?

- A. 8 kHz
- B. 10 kHz
- C. 16 kHz
- D. 6 kHz

Answer: C — $BW = 2 \times (5 + 3) = 16 \text{ kHz}$



Checkpoint Quiz 4 / 10

What do constellation diagrams display?

- A. Antenna radiation patterns
- B. Phase and amplitude states of a digital signal
- C. Frequency vs. time
- D. Modulation index vs. audio frequency

Answer: B — I vs Q plot showing all possible signal states (amplitude + phase)



Checkpoint Quiz 5 / 10

What is the bit rate of a 1200-baud QPSK signal?

- A. 1200 bps
- B. 600 bps
- C. 2400 bps
- D. 4800 bps

Answer: C — QPSK: 2 bits/symbol; $1200 \text{ baud} \times 2 = 2400 \text{ bps}$



Checkpoint Quiz 6 / 10

What causes key clicks in CW transmission?

- A. Excessive power
- B. Oscillator chirp
- C. Rise/fall time too fast
- D. Antenna mismatch

Answer: C — Key clicks: keying rise/fall time too fast → wide sidebands that splatter across the band



Checkpoint Quiz 7 / 10

What is a primary advantage of spread spectrum?

- A. Narrower bandwidth
- B. Lower power
- C. Resistance to interference and jamming
- D. Better audio quality

Answer: C — Spreading signal across wide bandwidth makes it resistant to narrowband interference



Checkpoint Quiz 8 / 10

FEC allows the receiver to do what that ARQ cannot?

- A. Increase data rate
- B. Detect and correct errors without requesting retransmission
- C. Compress the signal
- D. Synchronize clock automatically

Answer: B — FEC corrects errors without asking for a retransmit; ARQ must request retransmission



Checkpoint Quiz 9 / 10

What material in some old RF transistors requires extreme caution to avoid creating dust?

- A. Lead solder
- B. Silicon dioxide
- C. Beryllium oxide (BeO)
- D. Tin plating

Answer: C — BeO dust is extremely toxic; never sand, grind, or break components containing it



Checkpoint Quiz 10 / 10

Your neighbors live near your antenna. Which MPE environment applies at their location?

- A. Controlled — they're near an amateur station
- B. Uncontrolled — general public, stricter limits apply
- C. No limits apply to neighboring properties
- D. Controlled if they have been notified

Answer: B — Uncontrolled environment (general public) stricter MPE limits apply at all neighboring locations



Before Module 7

Your assignment:

- Drill HamStudy E8A-E8D and E0A to 80%+ confidence
- Practice Carson's rule and the RMS voltage formula — both appear directly on the exam
- E0A safety: know BeO hazard, duty cycle effect on MPE, controlled vs uncontrolled

Coming up:

Module 7: Antennas and Transmission Lines — the final module. Real antenna math, Smith chart, transmission line theory. 8 questions tied with E7. Come with Module 3 math fresh.